

Assignment 4

Functions

- 1- Explain the difference between passing (**Value type parameters**) by value and by reference then write a suitable c# example.
- 2- Explain the difference between passing (**Reference type parameters**) by value and by reference then write a suitable c# example.
- 3- Write a c# Function that accept 4 parameters from user and return result of summation and subtracting of two numbers
- 4- Write a program in C# Sharp to create a function to calculate the sum of the individual digits of a given number.

Output should be like

Enter a number: 25

The sum of the digits of the number 25 is: 7

- 5- Create a function named "IsPrime", which receives an integer number and returns true if it is prime, or false if it is not:
- 6- Create a function named MinMaxArray, to return the minimum and maximum values stored in an array, using reference parameters
- 7- Create an iterative (non-recursive) function to calculate the factorial of the number specified as parameter
- 8- Create a function named "ChangeChar" to modify a letter in a certain position (0 based) of a string, replacing it with a different letter

Enum and Struct

1. Create an enum called "WeekDays" with the days of the week (Monday to Sunday) as its members. Then, write a C# program that prints out all the days of the week using this enum.
2. Define a struct "Person" with properties "Name" and "Age". Create an array of three "Person" objects and populate it with data. Then, write a C# program to display the details of all the persons in the array.

3. Create an enum called "Season" with the four seasons (Spring, Summer, Autumn, Winter) as its members. Write a C# program that takes a season name as input from the user and displays the corresponding month range for that season. Note range for seasons (spring march to may , summer june to august , autumn September to November , winter December to February)

4- Assign the following Permissions (Read, write, Delete, Execute) in a form of Enum.

- Create Variable from previous Enum to Add and Remove Permission from variable, check if specific Permission is existed inside variable

5. Create an enum called "Colors" with the basic colors (Red, Green, Blue) as its members. Write a C# program that takes a color name as input from the user and displays a message indicating whether the input color is a primary color or not.

6. Create a struct called "Point" to represent a 2D point with properties "X" and "Y". Write a C# program that takes two points as input from the user and calculates the distance between them.

7. Create a struct called "Person" with properties "Name" and "Age". Write a C# program that takes details of 3 persons as input from the user and displays the name and age of the oldest person.

